

MC score ____/61

FR score ____/99

Total Score ____/180

Lab score ____/20

Circuit Lab Div C.

Hawk and Hornet Invitational

Do not open this test until instructed by your proctor

- You have 50 Minutes to complete both this test and a lab module.
- You may split up to work on the test and lab separately if desired.
- You are allowed a graphing calculator.
- You may separate the pages of this test.
- Use correct units when necessary
- This event consists of three parts with 40 questions total:
- Part 1: Multiple Choice - 30 questions (60pts)
- Part 2: Free Response - 10 questions (99 pts)
- Part 3: Lab - (20 pts)

Team Name: _____

Team #: _____

Student Names:

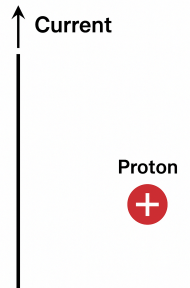
Multiple choice(2 pts each) - circle the correct answer

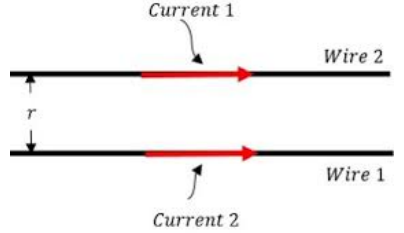
1. A 9V battery is connected to a 10 ohm resistor and a 5 ohm resistor in series.
What is the current through the 5 ohm resistor?
 - a) 2.7 A
 - b) 1.8 A
 - c) 0.6 A
 - d) 0.9 A
2. The current carrying capacity of a standard wire *most* depends on what?
 - a) Wire material
 - b) Gauge
 - c) Length
 - d) Insulation material
3. The voltage rating of a wire depends on what?
 - a) Wire material
 - b) Gauge
 - c) Length
 - d) Insulation material
4. Who invented the first electric battery? Bonus point if you can name it
 - a) Thomas Edison
 - b) Michael Faraday
 - c) Nikola Tesla
 - d) Alessandro Volta

Called a: _____

5. Which factors determine how dangerous a shock may be?
 - a) Voltage
 - b) Current
 - c) Frequency
 - d) Contact duration
 - e) All of the above

6. What can be used to produce a magnetic field? (circle all that apply)
- a) A very hot iron nail
 - b) A changing current
 - c) A steady-state current
 - d) A stationary proton
 - e) A moving neutron
7. What can be used to produce an electric field? (circle all that apply)
- a) A changing magnetic field
 - b) A stationary lone proton
 - c) A moving lone electron
 - d) A moving neutron
8. What is the peak to peak voltage of a standard American electrical outlet?
- a) 120V
 - b) 240V
 - c) 340V
 - d) 170V
9. If an electron has a velocity out of the page in a uniform magnetic field toward the left of the page. In what direction is the force experienced by the electron?
- a) Into the page
 - b) Towards the top of the page
 - c) Towards the bottom of the page
 - d) Towards the left of the page
10. A current flows through a wire towards the top of the page as shown. On the right of the wire is a stationary proton. In what direction is the Lorentz force exerted on the proton?
- a) Toward the wire
 - b) Away from the wire
 - c) No force exerted
 - d) Into the page



11. Approximately how efficient are typical LEDs?
- 30%
 - 90%
 - 5%
 - 80%
12. A 5 ohm resistor and two 10 ohm resistors are all placed in parallel. What is the equivalent resistance of the circuit?
- 2.6 ohms
 - 2.5 ohms
 - 4 ohms
 - 8.3 ohms
13. Two side by side wires both carry current in the same direction as shown. What direction is the force on each wire?
- Toward the other wire
 - Away from the other wire
 - No force exerted
 - One goes into the page and the other goes out of the page
- 
14. Why are there no magnetic monopoles?
- Valence electron interference
 - Quantized electron energies
 - Trick question, there are magnetic monopoles
 - That's just the way the universe works
15. When a capacitor fails due to overvoltage, what is it that fails?
- The plates touch due to electrostatic attraction
 - The dielectric undergoes dielectric breakdown
 - Electrons flow through the insulative package
 - The dielectric inside reacts with the metal plates
16. A point charge q_1 has a magnitude of 9nC . A second charge q_2 has a magnitude of -1.5nC and is located 0.12m from the first charge. Determine the net force each charge exerts on the other.
- $1.2\text{E}-16\text{ N}$
 - $1.1\text{E}-16\text{ N}$
 - $6.2\text{E}-8\text{ N}$
 - $8.4\text{E}-6\text{ N}$

17. Magnetism is _____
- a) A fundamental force separate from the electric force
 - b) Only present in transition metals
 - c) The result of relativistic length contraction
 - d) Magic
18. For any two resistors, which of the following is **false**?
- a) The equivalent series resistance will always be the sum of the individual resistances
 - b) The equivalent parallel resistance will always be greater than or equal to the largest resistor
 - c) The equivalent parallel resistance will always be less than or equal to the smallest resistor
 - d) The resistors in series will form a voltage divider
19. A Van De Graaff Generator can produce long electrical arcs but can be touched safely. Why is this the case?
- a) They have high current but low voltage
 - b) They have high voltage but low current
 - c) They have very low frequency
 - d) They have very high frequency
20. How fast do electrons move through a typical DC current-carrying wire?
- a) Exactly the speed of light
 - b) Almost (99.9999%) the speed of light
 - c) A few millimeters per second
 - d) Electrons don't actually move in a wire
21. Which of the following figures did not contribute to Maxwell's equations?
- a) Heinrich Hertz
 - b) Carl Friedrich Gauss
 - c) André-Marie Ampere
 - d) Michael Faraday

22. 50mA of current flows into a 2F capacitor for 25 seconds before it is disconnected from the current source. About how much energy is stored in the capacitor after it has been disconnected?
- a) 2.5 J
 - b) 0.6 J
 - c) 1.2 J
 - d) 0.4 J
23. Which type of capacitor is typically polarized?
- a) Electrolytic
 - b) Ceramic
 - c) Film
 - d) SMT
24. Which of the following is **not** an equation for power?
- a) VI
 - b) V^2/R
 - c) I^2R
 - d) V^2R
25. A device with a large load (like a refrigerator) acts most similar to
- a) A resistor with a high resistance
 - b) A resistor with a low resistance
 - c) A DC voltage source
 - d) A diode
26. Three diodes ($V_f = 0.7V$) and a 10 ohm resistor are placed in series and are connected to a battery with voltage V_s . Which equation best represents the current in the circuit?
- a) $(V_s - 0.7)/10$
 - b) $(V_s - 10 - 0.7)$
 - c) $(V_s - (3 * 0.7))/10$
 - d) $(V_s/10) - (3 * 0.7)$
27. What happens if a magnet is broken in half?
- a) One half becomes a north pole and the other half a south pole
 - b) The magnet stops working
 - c) It splits into two smaller magnets
 - d) The universe explodes

28. Is light a particle or a wave?
- a) Light is a particle because photons are discrete packets of energy
 - b) Light is a wave because it interferes with itself
 - c) Light is either a particle or a wave depending on its frequency
 - d) Light is both a particle and a wave
29. Which of the following would make a resistor have a higher resistance?
- a) Using a more conductive material
 - b) Heating up the resistor
 - c) Increasing cross-sectional area
 - d) Decreasing length
30. How hard was this test? (All answers are correct)
- a) Easy
 - b) Medium
 - c) Hard
 - d) Just right

Free Response - You may round all answers to the nearest tenth

1. a) You have a red LED ($V_f = 1.8\text{V}$) that you want to run at 35mW using a 3.7V battery. What value resistor must be used in series to achieve this? (5 pts)

b) Find the power dropped across the resistor (3pts)

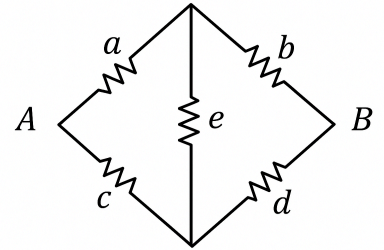
c) Find the number of LEDs you could put in parallel before burning out the resistor (assuming it is a typical $1/4\text{W}$ resistor) (3pts)

2. In a sentence or two, what is the difference between a buck converter and a boost converter? (In terms of what they do, not how they work) (3pts)

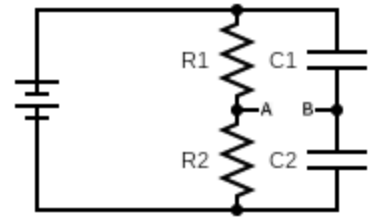
3. What is the resonant frequency, in Hz, of an LC oscillator with an inductance of 700uH and a capacitance of 8mF (5pts)
4. Write the units for the following values using only Coulombs(C), Joules(J), Seconds(s). (3 pts each)
- a) Voltage
 - b) Current
 - c) Capacitance
 - d) Power
 - e) Impedance

5. List the following materials in order of descending conductance:
Tap Water, Copper, Gold, Distilled Water, Silver (5pts)
6. You have a blue LED ($V_f = 3.0V$) and a voltage source with variable voltage V_s . Assuming you have a $1/4W$ variable resistor in series that always makes the current through the LED $20mA$, what is the largest voltage V_s can be without destroying your resistor? (5pts)

7. Derive a general solution for the equivalent resistance between nodes A and B of this resistor network (25 pts)

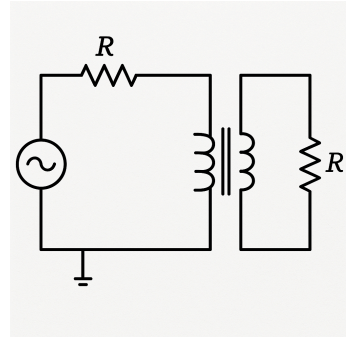


8. a) In the image shown, a 9V battery is connected to two series resistors and two series capacitors. $R_1 = 5\ \Omega$ $R_2 = 10\ \Omega$ $C_1 = 3\text{nF}$ $C_2 = 8\text{nF}$, find the potential difference between points A and B. (5pts)



- b) find the energy stored in C_2 (5pts)

9. Adia builds the circuit shown using a 1:10 step up transformer. Both resistors shown have a value of 10 ohms and the AC source has a voltage of 10V RMS.



- 1) Adia calculates that the RMS current in the primary circuit is $10\text{V}/10\text{ohms} = 1\text{A}$
- 2) Thus the total power coming from the source is $10\text{V} \cdot 1\text{A} = 10\text{W}$.
- 3) On the secondary coil however, adia measures 100V
- 4) She calculates that there must be a current of $100\text{V}/10\text{ohms} = 10\text{A}$
- 5) which implies that the power on the secondary side of the circuit must be $100\text{V} \cdot 10\text{A} = 1000\text{W}$.
- 6) This cannot be true since for any transformer, power in = power out.

On what line was the first mistake Adia made in her calculations and why was it wrong?
(8 pts)

10. Draw the schematic symbol for the following components: (3pts each)

a) Relay

b) NPN BJT

c) Potentiometer

d) LED